



Stomach and Duodenum — MCQ Practice

Physiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. Which gastric cell type secretes both hydrochloric acid and intrinsic factor?

EASY

- A. Chief cell
- B. Mucous neck cell
- C. Parietal cell
- D. G cell
- E. D cell

2. Which phase of gastric acid secretion contributes the largest share of total acid output?

EASY

- A. Cephalic phase
- B. Gastric phase
- C. Intestinal phase
- D. All three contribute equally
- E. None — acid secretion is constant regardless of phase

3. Which hormone directly stimulates gastric acid secretion by parietal cells?

EASY

- A. Somatostatin
- B. Secretin
- C. Gastrin
- D. Cholecystokinin
- E. Vasoactive intestinal peptide (VIP)

4. Which stomach region is thin-walled and specialized primarily for storing food rather than mixing it?

EASY

- A. Antrum
- B. Pylorus
- C. Fundus/body
- D. Cardia only
- E. Duodenal bulb

5. A patient with autoimmune destruction of gastric parietal cells (pernicious anemia) develops megaloblastic anemia over several years. Which mechanism directly links parietal cell loss to this anemia?

MEDIUM

- A. Loss of pepsinogen secretion impairs protein digestion, causing anemia
- B. Loss of intrinsic factor secretion impairs vitamin B12 absorption in the ileum
- C. Loss of gastrin secretion directly causes bone marrow suppression
- D. Loss of mucus secretion causes chronic gastric bleeding
- E. Loss of somatostatin secretion increases red cell destruction

6. A new drug mimics somatostatin's action on the stomach. What would be its expected net effect on gastric acid secretion?

MEDIUM

- A. Increased acid secretion via direct parietal cell stimulation
- B. Decreased acid secretion, since somatostatin is the master inhibitory hormone of gastric secretion
- C. No effect on acid secretion, only on motility
- D. Selectively increased pepsinogen secretion with no effect on acid
- E. Increased gastrin release with no downstream acid effect

7. A patient undergoes truncal vagotomy as part of an older ulcer surgery. Which phase of gastric acid secretion would be most directly and completely abolished?

MEDIUM

- A. Gastric phase, since it depends entirely on local distension
- B. Cephalic phase, since it is entirely vagally mediated
- C. Intestinal phase, since it depends on vagal reflexes exclusively
- D. None of the phases depend on vagal input
- E. All three phases equally, since acid secretion is purely hormonal

8. A patient with a gastrinoma (gastrin-secreting tumor, Zollinger-Ellison syndrome) develops refractory peptic ulcers. Which normal regulatory mechanism is being pathologically overridden?

MEDIUM

- A. The normal inhibition of gastrin release by low antral pH (below pH 3) and by somatostatin
- B. The normal absence of any gastric acid regulation
- C. The normal role of secretin in stimulating acid secretion
- D. The normal function of chief cells in acid secretion
- E. The normal role of CCK in promoting acid secretion

9. A bariatric surgeon performs a procedure that bypasses the duodenum, routing food from a small gastric pouch directly to the jejunum. What is the primary physiological rationale for excluding the duodenum specifically?

MEDIUM

- A. The duodenum secretes most of the body's digestive enzymes
- B. The duodenum absorbs even more nutrients than the stomach, so bypassing it reduces nutrient absorption
- C. The duodenum is the main site of gastric acid production
- D. The duodenum has no absorptive function, so bypassing it has purely mechanical benefits
- E. The duodenum is bypassed only to reduce gastrin release

10. A patient with chronic atrophic gastritis has reduced numbers of ECL cells. Which downstream effect on acid secretion is most directly expected?

MEDIUM

- A. Reduced histamine release, and therefore reduced direct stimulation of parietal cell acid secretion
- B. Increased pepsinogen secretion only
- C. No effect, since ECL cells are unrelated to acid secretion
- D. Increased somatostatin release, unrelated to ECL loss
- E. Increased intrinsic factor secretion

11. A meal high in protein is ingested. Which sequence best describes how amino acids in the stomach lumen increase acid secretion during the gastric phase?

MEDIUM

- A. Amino acids directly diffuse into parietal cells and trigger acid release
- B. Amino acids/peptides stimulate G cells to release gastrin, which then stimulates parietal cells (plus potentiates histamine)
- C. Amino acids inhibit D cells, indirectly raising acid output with no gastrin involvement
- D. Amino acids stimulate secretin release, which raises acid output
- E. Amino acids have no effect on acid secretion, only on pepsinogen

12. A patient with a duodenal ulcer has increased mucosal acid exposure. Considering gastric HCl's multiple functions, which of the following is NOT one of the physiological roles of gastric HCl described in this lecture?

MEDIUM

- A. Activating pepsinogen to pepsin
- B. Killing ingested bacteria
- C. Helping iron and calcium absorption
- D. Directly emulsifying dietary fat into micelles
- E. Promoting pancreatic and biliary secretion downstream

13. A patient has a lesion selectively destroying antral G cells while sparing fundic parietal and ECL cells. Which phase of acid secretion would be most directly impaired?

MEDIUM

- A. Cephalic phase only
- B. Gastric phase, since it depends heavily on gastrin release from the antrum
- C. Intestinal phase exclusively
- D. No phase would be affected, since G cells are not involved in acid regulation
- E. All phases equally, since G cells control the entire cephalic-vagal reflex

14. A patient reports minimal weight loss despite prolonged fasting-related nutrient malabsorption concerns and is found to have normal gastric emptying and normal absorption. Which statement best reflects the stomach's normal absorptive role?

MEDIUM

- A. The stomach is the primary site of nutrient absorption in the GI tract
- B. The stomach absorbs almost no nutrients, though it does absorb water, some ions, drugs, and alcohol
- C. The stomach absorbs all fat-soluble vitamins
- D. The stomach is responsible for most protein absorption
- E. The stomach absorbs glucose efficiently via sodium co-transport

15. A patient with Zollinger-Ellison syndrome (gastrinoma) has chronically elevated gastrin, but their D cells and somatostatin production are entirely normal. Why does somatostatin fail to adequately control the hypersecretion in this condition?

HARD

- A. Somatostatin has no physiological role in regulating gastrin at all
- B. The tumor autonomously secretes gastrin independent of the normal negative feedback loop that somatostatin exploits (via pH and paracrine signaling), so somatostatin cannot suppress a tumor that isn't responding to the normal regulatory circuit
- C. Somatostatin only inhibits histamine, not gastrin, so it was never expected to help
- D. Excess gastrin directly destroys D cells over time, always
- E. Somatostatin actually stimulates gastrinomas

16. A researcher blocks histamine H2 receptors on parietal cells with a specific antagonist. Gastrin and ACh pathways remain fully intact. Predict the effect on acid secretion, considering histamine's described role as potentiating other stimuli.

HARD

- A. Acid secretion would be completely abolished, since histamine is the sole acid stimulus
- B. Acid secretion would be substantially reduced but not entirely abolished, since gastrin and ACh have some direct stimulatory action on parietal cells independent of histamine potentiation, though much of their effect is normally amplified through histamine
- C. Acid secretion would be unaffected, since histamine plays no role in acid regulation
- D. Acid secretion would paradoxically increase
- E. This would only affect pepsinogen secretion, not acid

17. A patient has a vagotomy performed along with antrectomy (surgical removal of the gastrin-producing antrum) for severe peptic ulcer disease. Compared with vagotomy alone, why would this combined procedure produce a more profound reduction in acid secretion?

HARD

- A. Because it removes both the cephalic-phase vagal drive AND the gastric-phase gastrin drive, eliminating two of the three major stimulatory phases rather than just one
- B. Because antrectomy alone has no additional effect beyond vagotomy
- C. Because removing the antrum eliminates parietal cells directly
- D. Because vagotomy alone already abolishes all three phases, making antrectomy redundant
- E. Because antrectomy specifically targets the intestinal phase only

18. A patient with chronic H. pylori infection has predominantly antral-restricted disease, causing G-cell hyperplasia and hypergastrinemia while sparing the fundic parietal cell mass. How does this pattern typically differ from H. pylori infection that progresses to pan-gastritis with parietal cell atrophy?

HARD

- A. Both patterns produce identical acid secretion profiles
- B. Antral-predominant disease tends toward increased acid secretion and duodenal ulcer risk (via gastrin drive on an intact parietal mass), whereas pan-gastritis with parietal cell loss tends toward reduced acid secretion and gastric ulcer/cancer risk
- C. Antral-predominant disease always leads to achlorhydria
- D. Pan-gastritis always increases acid secretion further
- E. Neither pattern has any relationship to acid secretion physiology

19. A patient is found to have delayed gastric emptying of solids but normal emptying of liquids. Correlating this with the regional motility physiology of the stomach, which explanation is most consistent?

HARD

- A. Liquids and solids are handled identically by gastric motility, so this pattern is physiologically implausible
- B. The distal (antral) thick-walled, strongly contracting region — responsible for mechanically breaking down and expelling solids — is more likely impaired, while proximal storage/receptive relaxation (which governs liquid emptying more passively) is relatively preserved
- C. This pattern indicates a purely esophageal problem
- D. Liquid emptying depends entirely on the antrum, so this pattern is inconsistent with any recognized physiology
- E. This can only be explained by a pyloric stenosis affecting liquids selectively

20. A patient with long-standing diabetes develops gastroparesis with impaired vagal function. Considering that the cephalic phase is vagally mediated but the gastric phase involves both local (gastrin, distension) and vagal components, which best describes the expected acid secretion and motility profile?

HARD

- A. Only motility is affected; acid secretion by all three phases remains completely normal
- B. Both cephalic-phase acid secretion and vagally-mediated components of gastric motility/secretion would be blunted, while the local gastrin/distension-driven component of the gastric phase may partially persist
- C. Acid secretion increases paradoxically in all phases
- D. Vagal dysfunction has no bearing on either motility or secretion
- E. The intestinal phase becomes the dominant driver of all gastric motility

✓ Answer key

Q	Answer	Difficulty	Why
1	C	EASY	The parietal cell secretes both HCl and intrinsic factor.
2	B	EASY	The gastric phase contributes 60–70% of total acid output, the largest share.
3	C	EASY	Gastrin is the primary "on switch" hormone stimulating parietal cell acid secretion.
4	C	EASY	The proximal stomach (fundus and body) is thin-walled with weak, infrequent contractions specialized for storage.
5	B	MEDIUM	Parietal cells make intrinsic factor as well as HCl; its loss impairs B12 absorption in the ileum, causing megaloblastic anemia.
6	B	MEDIUM	Somatostatin is the master "off switch," inhibiting gastrin, histamine, and acid secretion broadly.
7	B	MEDIUM	The cephalic phase (sight/smell/taste-triggered) is entirely vagally mediated via ACh release onto parietal cells, so vagotomy abolishes it most directly.
8	A	MEDIUM	Normally, gastrin release is inhibited when antral pH drops below 3 and by somatostatin; autonomous tumor secretion overrides this negative feedback, driving unchecked acid secretion.
9	B	MEDIUM	The duodenum absorbs substantial nutrients (more than the stomach), so its surgical exclusion in bariatric procedures reduces nutrient absorption to promote weight loss.
10	A	MEDIUM	ECL cells secrete histamine, which directly stimulates parietal cell acid secretion; their loss reduces this histamine-driven pathway.
11	B	MEDIUM	Amino acids and protein digestion products in the gastric lumen stimulate G cells to release gastrin, which stimulates acid secretion directly and by potentiating histamine.
12	D	MEDIUM	

Fat emulsification is performed by bile salts, not gastric HCl; HCl activates pepsinogen, kills bacteria, aids $\text{Fe}^{2+}/\text{Ca}^{2+}$ absorption, and promotes downstream pancreatic/biliary/intestinal secretion.

13	B	MEDIUM	G cells (located in the antrum) drive much of the gastrin-mediated gastric phase; their loss would blunt this phase specifically, though direct vagal/ACh and histamine pathways would remain partially active.
14	B	MEDIUM	The stomach has very limited absorptive capacity for nutrients — its absorption is limited mainly to water, some ions, drugs, and alcohol.
15	B	HARD	Somatostatin normally suppresses gastrin release from G cells as part of a physiological feedback loop; a gastrin-secreting tumor operates autonomously outside this regulatory circuit, so somatostatin's normal paracrine inhibition is insufficient to control it.
16	B	HARD	Histamine directly stimulates parietal cells and also potentiates gastrin/ACh-driven secretion; blocking H2 receptors removes both the direct histamine effect and much of the amplification of the other pathways, substantially (though not completely) reducing acid secretion — the physiological basis for H2 blocker drugs.
17	A	HARD	Vagotomy removes cephalic-phase (and part of gastric-phase vagal) drive; antrectomy removes the antral G-cell mass responsible for gastrin-driven gastric-phase secretion — combining them removes two major stimulatory pathways rather than one.
18	B	HARD	Antral-predominant <i>H. pylori</i> gastritis drives gastrin-mediated hyperacidity (favoring duodenal ulcers) because the acid-producing fundic mucosa is preserved, while pan-gastritis with parietal cell loss/atrophy reduces acid output and favors gastric ulcer/cancer risk — a classic discriminator in gastric physiology and pathology.
19	B	HARD	Solid emptying depends heavily on strong antral peristalsis to grind and sieve particles to a small enough size to pass the pylorus, while liquid emptying is more dependent on the pressure gradient from proximal stomach tone; selective solid-emptying delay points toward antral motor dysfunction rather than global gastric failure.
20	B	HARD	Since the cephalic phase is entirely vagal and the gastric phase has both a vagal component and a local (gastrin/distension) component, vagal dysfunction (as in diabetic autonomic neuropathy) would blunt cephalic-phase secretion and vagally-mediated motility/secretion, while local gastric-phase mechanisms may partially compensate — consistent with the mixed picture seen in diabetic gastroparesis.